

• Notes of Lecture 2

§ 2.1 The Marcinkiewicz interpolation theorem

2.1.1 Preliminaries

Let $(X, \mathcal{A}, \mu)$ be a measure space. For $0 < q < r \leq \infty$, define

$$L^q(X) + L^r(X) := \left\{ f \text{ } \mathcal{A}\text{-measurable: } f = g + h \text{ for some } g \in L^q(X) \text{ and } h \in L^r(X) \right\}.$$

Lemma 1. If $(X, \mathcal{A}, \mu)$ is a measure space and $0 < q < p < r \leq \infty$, then

$$L^p(X) \subseteq L^q(X) + L^r(X).$$

Proof. Let $f \in L^p(X)$ be given. For $\alpha > 0$, write

$f = \underbrace{f \cdot \chi_{\{|f| > \alpha\}}}_{f^\alpha} + \underbrace{f \cdot \chi_{\{|f| \leq \alpha\}}}_{f_\alpha}$. Here, χ_A denotes the characteristic function of a set $A \subseteq X$; $\chi_A(x) = \begin{cases} 1, & \text{if } x \in A \\ 0, & \text{if } x \in X \setminus A \end{cases}$.

We claim that $f^\alpha \in L^q(X)$ and $f_\alpha \in L^r(X)$. Indeed,

$$\begin{aligned} \int_X |f^\alpha(x)|^q d\mu(x) &= \int_{\{|f| > \alpha\}} |f(x)|^q d\mu(x) = \int_{\{|f| > \alpha\}} |f(x)|^p \cdot |f(x)|^{q-p} d\mu(x) \\ &\leq \alpha^{q-p} \int_{\{|f| > \alpha\}} |f(x)|^p d\mu(x) \\ &\leq \alpha^{q-p} \int_X |f(x)|^p d\mu(x) < \infty \end{aligned}$$

and

$$\begin{aligned} \int_X |f_\alpha(x)|^r d\mu(x) &= \int_{\{|f| \leq \alpha\}} |f(x)|^r d\mu(x) = \int_{\{|f| \leq \alpha\}} |f(x)|^p |f(x)|^{r-p} d\mu(x) \\ &\leq \alpha^{r-p} \int_{\{|f| \leq \alpha\}} |f(x)|^p d\mu(x) \\ &\leq \alpha^{r-p} \int_X |f(x)|^p d\mu(x) < \infty. \end{aligned}$$

So, $f \in L^q(X) + L^r(X)$.

Exercise. If $1 \leq q < r \leq \infty$, prove that

$$L^q(X) + L^r(X) \subseteq L^1_{loc}(X).$$

Definition 2. Let $(X, \mathcal{A}, \mu)$ and $(Y, \mathcal{B}, \nu)$ be two measure spaces. Let T be an operator defined on a linear space V of complex-valued $\mathcal{A}$ -measurable functions that maps functions in V to complex-valued $\mathcal{B}$ -measurable functions. We say that T is:

- **linear** if for all $\mathcal{A}$ -measurable functions $f, g \in V$ and $c \in \mathbb{C}$ one has
 $T(f+g) = T(f) + T(g)$, $T(cf) = cT(f)$.

- **sublinear** if for all $\mathcal{A}$ -measurable functions $f, g \in V$ and $c \in \mathbb{C}$ one has
 $|T(f+g)| \leq |T(f)| + |T(g)|$, $|T(cf)| = |c| |T(f)|$.

Definition 3. Let $(X, \mathcal{A}, \mu)$ and $(Y, \mathcal{B}, \nu)$ be two measure spaces and let $0 < p < \infty$.

- If T is an operator such that

$$\|T(f)\|_{L^p(Y)} \leq C_p \|f\|_{L^p(X)}$$

for some absolute constant $C_p > 0$, then T is of **strong-type** (p, p) .

- If T is an operator such that

$$\nu(\{y \in Y: |T(f)(y)| > a\}) \leq \frac{C_p}{a^p} \int_X |f(x)|^p d\mu(x)$$

for all $a > 0$, with $C_p > 0$ being an absolute constant, then T is of **weak-type** (p, p) .

Remark. If T is of strong-type (p, p) then T is of weak-type (p, p) . Indeed, for $\alpha > 0$, we have

$$\begin{aligned} C_p \|f\|_{L^p(X)}^p &\geq \int_Y |T(f)(y)|^p dv(y) \geq \int_{\{|T(f)| > \alpha\}} |T(f)(y)|^p dv(y) \\ &\geq \alpha^p v(\{y \in Y: |T(f)(y)| > \alpha\}). \end{aligned}$$

2.1.2 The Marcinkiewicz interpolation theorem

Theorem 4. Let (X, A, μ) and (Y, B, ν) be two σ -finite measure spaces.

Let $0 < q < r \leq \infty$. If T is a sublinear operator defined on $L^q(X) + L^r(X)$ such that:

- T is of weak-type (q, q) ,
- T is of weak-type (r, r) if $r < \infty$ and if $r = \infty$ T is of strong-type (∞, ∞) ,

then T is of strong-type (p, p) for all $p \in (q, r)$.

Proof. Case 1 ' $r < \infty$.' Fix a $p \in (q, r)$ and take an $f \in L^p(X)$. We have

$$\begin{aligned} \|T(f)\|_{L^p(Y)}^p &= \int_Y |T(f)(y)|^p dv(y) \\ &= \int_Y \left(\int_0^{|T(f)(y)|} p \cdot \alpha^{p-1} d\alpha \right) dv(y) \\ &= \int_0^\infty p \alpha^{p-1} v(\{y \in Y: |T(f)(y)| > \alpha\}) d\alpha \end{aligned}$$

For $a > 0$, we decompose $f = \underbrace{f \cdot \chi_{\{|f| > a\}}}_{f^a} + \underbrace{f \cdot \chi_{\{|f| \leq a\}}}_{f_a}$.

Then, as T is sublinear,

$$|T(f)(y)| \leq |T(f^a)(y)| + |T(f_a)(y)|$$

and so,

$$\begin{aligned} \{y \in Y: |T(f)(y)| > a\} &\subseteq \{y \in Y: |T(f^a)(y)| > a/2\} \\ &\quad \cup \{y \in Y: |T(f_a)(y)| > a/2\}. \end{aligned}$$

Hence,

$$\|T(f)\|_{L^p(Y)}^p \leq A + B, \text{ where}$$

$$A := \int_0^\infty p a^{p-1} v(\{y \in Y: |T(f^a)(y)| > a/2\}) da$$

and

$$B := \int_0^\infty p a^{p-1} v(\{y \in Y: |T(f_a)(y)| > a/2\}) da.$$

We shall handle each term separately.

To bound A , we use the fact that T is of weak-type (q, q) ,

$$A \leq \int_0^\infty p a^{p-1} \left(\frac{C_q}{(a/2)^q} \int_X |f^a(x)|^q d\mu(x) \right) da$$

$$= 2^q p C_q \int_0^\infty a^{p-q-1} \left(\int_{\{|f| > a\}} |f(x)|^q d\mu(x) \right) da$$

$$= 2^q p C_q \int_X |f(x)|^q \left(\int_0^{|f(x)|} a^{p-q-1} da \right) d\mu(x)$$

$$= 2^q p C_q \int_X |f(x)|^q \frac{|f(x)|^{p-q}}{p-q} d\mu(x)$$

$$= \frac{2^q p C_q}{p-q} \int_X |f(x)|^p d\mu(x).$$

To bound B , we use the fact that T is of weak-type (r, r) ,

$$\begin{aligned}
 B &\leq \int_0^\infty p \alpha^{p-1} \left(\frac{C_r}{(\alpha/2)^r} \int_X |f_\alpha(x)|^r d\mu(x) \right) d\alpha \\
 &= 2^r p C_r \int_0^\infty \alpha^{p-r-1} \left(\int_{\{|f| \leq \alpha\}} |f(x)|^r d\mu(x) \right) d\alpha \\
 &= 2^r p C_r \int_X |f(x)|^r \left(\int_{|f(x)|}^\infty \alpha^{p-r-1} d\alpha \right) d\mu(x) \\
 &= 2^r p C_r \int_X |f(x)|^r \frac{|f(x)|^{p-r}}{r-p} d\mu(x) \\
 &= \frac{2^r p C_r}{r-p} \int_X |f(x)|^p d\mu(x)
 \end{aligned}$$

Hence, $\|T(f)\|_{L^p(Y)} \leq \left(\frac{2^q C_q}{p-q} + \frac{2^r C_r}{r-p} \right)^{1/p} p^{1/p} \|f\|_{L^p(X)}$,

where C_q and C_r are the constants in the weak-type (q, q) and (r, r) inequalities, respectively.

Case 2 ' $r = \infty$ '. In this case we assume that there exists a constant $C_\infty > 0$ such that

$$\|T(f)\|_{L^\infty(Y)} \leq C_\infty \|f\|_{L^\infty(X)}.$$

Fix a $p \in (q, \infty)$ and take an $f \in L^p(X)$.

For $\alpha > 0$, decompose $f = \underbrace{f \cdot \chi_{\{|f| > \frac{\alpha}{2C_\infty}\}}}_{\tilde{f}^\alpha} + \underbrace{f \cdot \chi_{\{|f| \leq \frac{\alpha}{2C_\infty}\}}}_{\tilde{f}_\alpha}.$

We have

$$|T(f)| \leq |T(\tilde{f}^\alpha)| + |T(\tilde{f}_\alpha)|$$

and so,

$$\begin{aligned}
 \nu(\{y \in Y: |T(f)(y)| > \alpha\}) &\leq \nu(\{y \in Y: |T(\tilde{f}^\alpha)(y)| > \alpha/2\}) \\
 &\quad + \nu(\{y \in Y: |T(\tilde{f}_\alpha)(y)| > \alpha/2\}).
 \end{aligned}$$

Note that since $\|\tilde{f}_\alpha\|_{L^\infty(Y)} \leq \frac{\alpha}{2C_\infty}$,

one has $\|T(\tilde{f}_a)\|_{L^\infty(Y)} \leq C_\infty \cdot \frac{a}{2C_\infty} = a/2$ and so,

$$v(\{y \in Y: |T(\tilde{f}_a)(y)| > a/2\}) = 0.$$

Hence,

$$v(\{y \in Y: |T(f)(y)| > a\}) \leq v(\{y \in Y: |T(\tilde{f}^a)(y)| > a/2\}).$$

We have

$$\begin{aligned} \|T(f)\|_{L^p(Y)}^p &= \int_0^\infty p a^{p-1} v(\{y \in Y: |T(f)(y)| > a\}) da \\ &\leq \int_0^\infty p a^{p-1} v(\{y \in Y: |T(\tilde{f}^a)(y)| > a/2\}) da \\ &\leq \int_0^\infty p a^{p-1} \left(\frac{C_q}{(a/2)^q} \int_X |\tilde{f}^a(x)|^q d\mu(x) \right) da \\ &= 2^q C_q p \int_0^\infty a^{p-q-1} \left(\int_{\{|f| > a/2C_\infty\}} |f(x)|^q d\mu(x) \right) da \\ &= 2^q C_q p \int_X |f(x)|^q \left(\int_0^{2C_\infty |f(x)|} a^{p-q-1} da \right) d\mu(x) \\ &= 2^q C_q p \int_X |f(x)|^q \cdot \frac{(2C_\infty)^{p-q} |f(x)|^{p-q}}{p-q} d\mu(x) \\ &= \frac{2^q C_q p (2C_\infty)^{p-q}}{p-q} \int_X |f(x)|^p d\mu(x). \quad \blacksquare \end{aligned}$$

Corollary 5. The non-centred maximal operator M satisfies

$$\|M(f)\|_{L^p(\mathbb{R}^d)} \leq \frac{2p \cdot 3^{d/p}}{p-1} \|f\|_{L^p(\mathbb{R}^d)}$$

for all $1 < p < \infty$.

It follows from the exercise that there exists a sequence $(r_k)_{k \in \mathbb{N}}$ of non-negative real numbers such that $r_k \xrightarrow[k \rightarrow \infty]{} 0^+$ and $f(x) = \lim_{k \rightarrow \infty} f_{r_k}(x)$ for a.e. $x \in \mathbb{R}^d$.

So, to complete the proof, it suffices to prove that the limit $\lim_{r \rightarrow 0} f_r(x)$ exists for a.e. $x \in \mathbb{R}^d$ (as in such a case it must be equal to $f(x)$).

To this end, consider the oscillation $\underline{O}(f)$ of f as follows

$$\underline{O}(f)(x) := \left| \limsup_{r \rightarrow 0^+} f_r(x) - \liminf_{r \rightarrow 0^+} f_r(x) \right|, \quad x \in \mathbb{R}^d.$$

Our goal is to show that $\underline{O}(f) = 0$ a.e. on $\mathbb{R}^d$ or, equivalently,

$$|\{x \in \mathbb{R}^d : \underline{O}(f)(x) > 0\}| = 0. \quad (1)$$

Exercise. Show that if $h \in C_c(\mathbb{R}^d)$ (that is, h is continuous with compact support) and non-negative then $\underline{O}(h) = 0$ on $\mathbb{R}^d$.

Let $\varepsilon > 0$ be given. Since $C_c(\mathbb{R}^d)$ is dense in $L^1(\mathbb{R}^d)$, there exists a non-negative $h \in C_c(\mathbb{R}^d)$ such that

$$\|f - h\|_{L^1(\mathbb{R}^d)} < \varepsilon^2 / 2 \cdot 3^d. \quad (2)$$

Since $\underline{O}(f) \leq \underline{O}(f-h) + \underline{O}(h)$, it follows from the exercise above (and the fact that $\underline{O}(f-h) \leq 2M(f-h)$) that

$$\underline{O}(f) \leq 2M(f-h). \quad (3)$$

We thus have

$$\begin{aligned} |\{x \in \mathbb{R}^d : \underline{O}(f)(x) > \varepsilon\}| &\stackrel{(3)}{\leq} |\{x \in \mathbb{R}^d : M(f-h)(x) > \varepsilon/2\}| \\ &\leq \frac{3^d}{(\varepsilon/2)} \int_{\mathbb{R}^d} |f(x) - h(x)|^2 dx \quad (M \text{ is of weak type (2,2)}) \\ &\stackrel{(2)}{\leq} \frac{2 \cdot 3^d}{\varepsilon} \varepsilon^2 / 2 \cdot 3^d = \varepsilon \end{aligned}$$

Since $|\{x \in \mathbb{R}^d : \underline{O}(f)(x) > 0\}| = \lim_{N \rightarrow \infty} |\{x \in \mathbb{R}^d : \underline{O}(f)(x) > 1/N\}|$,

we have

$$\begin{aligned} |\{x \in \mathbb{R}^d : \underline{O}(f)(x) > 0\}| &= \lim_{N \rightarrow \infty} |\{x \in \mathbb{R}^d : \underline{O}(f)(x) > 1/N\}| \\ &\leq \lim_{N \rightarrow \infty} N^{-1} = 0, \text{ which is (1).} \end{aligned}$$

Proof. In the previous lecture we showed that

- $|\{x \in \mathbb{R}^d : M(f)(x) > \alpha\}| \leq \frac{3^d}{\alpha} \|f\|_{L^1(\mathbb{R}^d)} , \alpha > 0$
- $\|M(f)\|_{L^\infty(\mathbb{R}^d)} \leq \|f\|_{L^\infty(\mathbb{R}^d)}$

Hence, Theorem 4 implies that M is of strong-type (p,p) for all $p \in (1, \infty)$ i.e. $\|M(f)\|_{L^p(\mathbb{R}^d)} \leq C_p \|f\|_{L^p(\mathbb{R}^d)}$ with $C_p \leq \left(\frac{2^d \cdot 3^d \cdot p \cdot (2 \cdot 1)^{p-1}}{p-1} \right)^{1/p} = \left(\frac{p}{p-1} \right)^{1/p} \cdot 2 \cdot 3^{d/p} \leq \frac{2p 3^{d/p}}{p-1}$. ■

Remark. Similar weak-type $(1,1)$ bounds and strong-type (p,p) bounds for $1 < p \leq \infty$ hold for the centred maximal operator M_c and for maximal operators taken with respect to cubes instead of balls.

§ 2.2 The Lebesgue differentiation theorem

As a corollary of the weak-type inequality for the maximal operator, we obtain the following extension of the fundamental theorem of Calculus.

Theorem 6. For $f \in L^1_{loc}(\mathbb{R}^d)$ one has

$$f(x) = \lim_{r \rightarrow 0^+} \frac{1}{|B(x,r)|} \int_{B(x,r)} f(y) dy$$

for a.e. $x \in \mathbb{R}^d$.

Proof. We assume first that $f \in L^1(\mathbb{R}^d)$. We may assume that f is non-negative.

For $r > 0$, define

$$f_r(x) := \frac{1}{|B(x,r)|} \int_{B(x,r)} f(y) dy , \quad x \in \mathbb{R}^d$$

Exercise. Show that $\lim_{r \rightarrow 0^+} \|f_r - f\|_{L^1(\mathbb{R}^d)} = 0$.

One may deduce the general $L^1_{loc}(\mathbb{R}^d)$ - case from the special case treated above (exercise). ■

Remark. In **Theorem 6** the centred balls can be replaced by non-centred balls (containing x) or cubes.

Corollary 7. For $f \in L^1_{loc}(\mathbb{R}^d)$ one has
 $|f(x)| \leq M(f)(x)$ for a.e. $x \in \mathbb{R}^d$.
Therefore, for all $p \in (1, \infty]$ there exists a constant $C_{p,d} > 0$ such that

$$\|f\|_{L^p(\mathbb{R}^d)} \leq \|M(f)\|_{L^p(\mathbb{R}^d)} \leq C_{p,d} \|f\|_{L^p(\mathbb{R}^d)}$$

The material of this lecture is based on the following references:

- Grafakos, Loukas. Classical Fourier Analysis. Third Edition. Graduate Texts in Mathematics, 249. Springer. New York, 2014, xviii + 638 pp
- Stein, Elias M. Singular integrals and differentiability properties of functions. Princeton Mathematical Series, No. 30. Princeton University Press, Princeton, N.J. 1970 xiv + 290 pp